hw6

Problem 1

1

$p_A(z) = z^m + a_{m-1} z^{m-1} + a_{m-2} z^{m-2} + \dots + a_1 z + a_0$

$a_{m-1} = -tr(A) \quad a_j \in \mathbb{R}, j = 0 \dots m-2$

Base case 2x2: $C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $P_C(z) = (z-a)(z-d) - bc =$

$z^2 - dz - az + ad - bc = z^2 - (d+a)z + (ab - bc)$

here we see $a_{n-1} = a_1 = -(d+a) = -tr(A)$ so we are good

Inductive hyp: True for $m-1$ WTS true for m

Induc Step:

Let B be an $m-1 \times m-1$ ~~poly~~ matrix so

$p_B(z) = z^{m-1} + a_{m-2} z^{m-2} + \dots + a_1 z + a_0 \quad a_{m-2} = -tr(B)$

Let $A = \begin{bmatrix} * & \cdots & * \\ \vdots & B & \\ * & & \end{bmatrix}$ then ~~$P_A(z) = (z - a_{11}) \det(B) + \sum_{i=2}^{m} (z-a$~~

$P_A(z) = (z - a_{11}) \det(B) + \sum_{i=2}^{m} a_{i1} \det(M_{i1}) (-1)^{i+1}$ where ~~M_{i1} is~~

M_{i1} is the matrix by ~~removing~~ deleting row i col 1

~~Since M_{i1} ... is $m-1 \times m-1$, we know by induction hypothesis it will look like $P_{M_{i1}}(z) = z^{m-1} + x_{m-2} z^{m-2} + \dots + x_1 z + x_0$ where $x_{m-2} = -tr(M_{i1})$. Since x_{m-2} is the coef of z^{m-2}, it has 0 effect~~

Since we delete row i + col 1 + $i \neq 1$, we are deleting 2 different elems on the main diagonal (a_{11} + a_{ii}) so $\deg(\det(M_{i1})) = m-2$ so it has no effect on $a_{m-1} z^{m-1}$.

Taking $(z - a_{11}) \det B$ we get $z^m + a_{m-2} z^{m-1} + \dots + z a_0 + a_{11} z^{m-1} + \dots$ discarding the higher order terms, we have $z^m + a_{m-2} z^{m-1} + a_{11} z^{m-1}$ $= z^m + (a_{m-2} - a_{11}) z^{m-1}$ since $a_{m-2} = -tr(B)$ by induc hyp, $a_{m-2} - a_{11} = -tr(B) - a_{11} = -tr(A)$ so we have proven correct for $m \times m$ matrix

Problem 2

2

a) Let $\lambda \in \mathbb{C}$ + $Ax = \lambda x$ for some $x \in \mathbb{R}^n$ + $\mu \in \mathbb{R}$
then $(A - \mu I)x = Ax - \mu I x = \lambda x - \mu I x = \lambda x - \mu x =$
$(\lambda - \mu)x$ so $\lambda - \mu$ is eval for $A - \mu I$

b) Let A be nonsing. If $\lambda = 0$ is an eval then $\exists x \neq \vec{0}$
where $x \in \mathbb{R}^n$ st $Ax = \lambda x = 0x = 0$ but not possible b/c
A nonsing so $\lambda \neq 0$
If $Ax = \lambda x$ then $x = \lambda A^{-1}x$ ~~so~~
$\Rightarrow A^{-1}x = \frac{1}{\lambda}x$ so λ^{-1} is eval of A^{-1}

c) $A = P^{-1}DP$ + all evals equal then $P(x) = (x-\lambda)^m$ so let x be an evec
~~$Ax = \lambda x = P^{-1}DPx$ $PAx = DPx = \lambda Px$~~
$A = P^{-1}DP$ if all evals the same then $A = P^{-1}\lambda I P = \lambda P^{-1}P =$
λI so $A = \lambda I$ which is diagonal matrix

d) ~~If $Ax = \lambda x$ then~~ $\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$
so A^T + A must have same evals

e) If $\lambda \in \mathbb{C}$ + eval of A. ~~Since $A \in \mathbb{R}^{m \times m}$, det(A) ∈ ℝ so~~ then
it is a root of $P(x)$ characteristic poly so $\bar{\lambda}$ must be a root as well so $\bar{\lambda}$ is an
eval. Let $Ax = \lambda x$ then $\overline{Ax} = \overline{\lambda x}$ so $A = \bar{A}$ b/c $A \in \mathbb{R}^{m \times m}$ so
$A\bar{x} = \bar{\lambda}\bar{x}$ so since $\bar{\lambda}$ is an eval, $\bar{x}$ is the corresp evec
so if (λ, x) is an epair so is $(\bar{\lambda}, \bar{x})$

Problem 3

$$\left[\begin{array}{cccc} a_{11} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{array}\right.$$

(3)

Know in Schur decomp $A = QUQ^T$ where U is upper triang & diag elems are diagonal

Since $A \in \mathbb{R}^{n\times n}$ $A^* = A^T = A$ so if λ is eval & $\lambda \neq 0$ then let x be the corresp evec $(x \neq \vec{0})$ so $Ax = \lambda x \Rightarrow x^* A^* = \bar{\lambda} x^* \Rightarrow x^* A = \bar{\lambda} x^*$ $\Rightarrow x^* A x = \bar{\lambda} x^* x \Rightarrow x^* \lambda x = \bar{\lambda} x^* x \Rightarrow \lambda x^* x = \bar{\lambda} x^* x$ since $x^* x \neq 0$ (b/c $x \neq 0$) $\lambda = \bar{\lambda}$ so evals are real. Then since $A^* = A$

$QUQ^* = Q^* U^* Q^* \Rightarrow U = U^*$. ~~This~~ Since U is upper triang. this means U is diagonal & b/c all diag elms are ~~real~~ evals (which are real) $U^* = U^T = U =$ diag call that D. Then

$A = QDQ^T$

Problem 4

(4) Let λ be the eval for $A+E$ w/ $x \neq \vec{0}$ as the corresp evec then

$(A+E)x = \lambda x \Rightarrow Ax + Ex = \lambda x \Rightarrow \lambda x - Ax = Ex \Rightarrow (\lambda I - A)x = Ex$ (*)

$\Rightarrow x = (\lambda I - A)^{-1} E x$ So $\|x\|_2 = \|(\lambda I - A)^{-1} E x\|_2 \le \|(\lambda I - A)^{-1}\|_2 \|E\|_2 \|x\|_2$

from (3) since $A = A^T$ $A = QDQ^T$ where Q orthogonal + D diagonal w/ evals on main diag so $\lambda I - A = \lambda Q I Q^T - QDQ^T = Q(\lambda I - D)Q^T$ so

$\|(\lambda I - D)^{-1}\|_2 = \|Q(\lambda I - D)^{-1}Q^T\|_2$ Since Q orthog, $= \|(\lambda I - D)^{-1}\|_2$

$= \frac{1}{\min_{\text{sing}}(\lambda I - D)}$ ← b/c diag $D = \begin{bmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_m\end{bmatrix}$ & $\lambda I = \begin{bmatrix}\lambda & & \\ & \ddots & \\ & & \lambda\end{bmatrix}$ $\lambda I - D = \begin{bmatrix}\lambda - \lambda_1 & & \\ & \ddots & \\ & & \lambda - \lambda_m\end{bmatrix}$

since it is diagonal, singular vals are abs val of evals so

$\|(\lambda I - D)^{-1}\|_2 = \frac{1}{|\lambda - \lambda_j|}$ where λ_j is st $\lambda - \lambda_j \le \lambda - \lambda_i$ for $i \in 1..m$

so back to (*) $\|x\|_2 \le \frac{1}{|\lambda - \lambda_j|} \|E\|_2 \|x\|_2 \Rightarrow$

$|\lambda - \lambda_j| \le \|E\|_2$

Note this proof assumes $\lambda \neq \lambda_j$. If $\lambda = \lambda_j$ then $|\lambda - \lambda_j| = 0$ + $\|E\|_2 \geq 0$ $\forall E$ so it is always true

Problem 5

Code:

```
A = [5 1 3; 1 2 1; 2 4 3];
H = [5 -7/sqrt(5) 1/sqrt(5); -sqrt(5) 24/5 13/5; 0 -2/5 1/5]
```

```
H_pred = hess(A)
err = abs(H - H_pred)
```

Output:

```
H =


    5.0000   -3.1305    0.4472
   -2.2361    4.8000    2.6000
         0   -0.4000    0.2000



H_pred =


    5.0000   -3.1305    0.4472
   -2.2361    4.8000    2.6000
         0   -0.4000    0.2000



err =


   1.0e-15 *


         0    0.4441    0.6661
         0         0    0.8882
         0    0.8327    0.0833
```

Problem 6

(6a) WTS $B = \hat{A} + w v_1^T + v_1 w^T \quad w = u + \delta v_1 \quad u = -\gamma \hat{A} v_1 \quad \delta = -\frac{1}{2}\gamma v_1^T u$ ← const

$B = F_1 \hat{A} F_1 = (I - \gamma v_1 v_1^T)\hat{A}(I - \gamma v_1 v_1^T) = (I - \gamma v_1 v_1^T)(\hat{A} - A\gamma v_1 v_1^T)$

$= \hat{A} - \hat{A}\gamma v_1 v_1^T - \gamma v_1 v_1^T \hat{A} + \gamma v_1 v_1^T \hat{A} \gamma v_1 v_1^T$ γ is a const so

$-\gamma v_1 v_1^T \hat{A} = -\gamma v_1 (\hat{A}^T v)^T = -\gamma v_1 (\hat{A} v)^T = v_1(\gamma \hat{A} v)^T = v_1 u^T$ so

$= \hat{A} + u v_1^T + v_1 u^T + \gamma v_1 v_1^T u v_1^T$

$\gamma v_1 v_1^T u v_1^T = v_1 \gamma v_1^T u v^T = v_1 2\delta v^T = v_1 2\delta v^T$ so

$= \hat{A} + u v_1^T + v_1 u^T + 2\delta v_1 v_1^T = \hat{A} + u v_1^T + \delta v_1 v_1^T + v_1 u^T + \delta v_1 v_1^T$

$= \hat{A} + (\underbrace{u + \delta v_1}_{w}) v_1^T + v_1 (u^T + \delta v_1^T) \qquad u^T + \delta v_1^T = (u + \delta v_1)^T = w^T$

$= \hat{A} + w v_1^T + v_1 w^T$ ☐

(b) Want to go from $A \to \begin{bmatrix} & \|x\| & 0 \\ & & B \end{bmatrix}$

To calc $B = A + w v_1^T + v_1 w^T$ since $v_1 w^T = (w v_1^T)^T$, only need to do it once

complexity of $w v_1^T$ is $(m-1)^2$ since both are $(m-1)$ vecs

& when calc w, we need u which is $\gamma A v$. Av is also $(m-1)^2$

& then mult by const is just $(m-1)$ so cost of u is $(m-1)^2 + (m-1)$

γ we calc once so it's just m (bk $m-1$ +1 for the 2) & so δ is $m+1$

($v^T u$ is $m-1$ & +1 for γ + 1 for $-\frac{1}{2}$). So total we have: $(m-1)^2 + (m-1)^2 + m - 1 + m + m + 1$

$\sim 2(m-1)^2$ ☐

(c) So for step 1 its $2(m-1)^2$ then step 2 its $2(m-2)^2$ so

$\sum_{k=1}^{m-2} 2(m-k)^2 = 2\sum_{k=1}^{m-2}(m-k)^2 = 2\sum_{k=1}^{m-1} k^2 = 2\frac{(m-1)(m)(2m-1)}{6} = \frac{1}{3}(2m^3 - 3m^2 + 1) \sim$

$\sim \frac{2}{3} m^3$ ☐